

What's Up with Duct Tape?

New tests for duct tape include testing the strapping that holds duct connections together and testing the longevity of duct sealants.

by **Iain Walker and Max Sherman**

It's been a couple of years since we last wrote about duct tape in the pages of *Home Energy*, and it is time to revisit this ever-popular subject. When last we left duct sealant durability issues, the Energy Performance of Buildings Group at Lawrence Berkeley National Laboratory (LBNL) had done an accelerated longevity test; we found that most everything worked except standard, cloth-backed rubber adhesive duct tape (see "Duct Lab Does Duct Tape," *HE* Sept/Oct '02, p. 13). In response, the state of California had limited the use of such tapes in new construction, and manufacturers were considering developing new products.

Several things have changed in the world of duct tape over the last couple of years. LBNL has completed another round of durability testing. There is a new version of Underwriters' Laboratories (UL) 181B that now includes testing the strapping that holds duct connections together. And there is a new American Society for Testing and Materials (ASTM) E2342-03 standard for testing the longevity of duct sealants.

Core-to-Collar Joints

Rather than the collar-to-plenum connection we tested previously ("Can Duct Tape Take the Heat?" *HE* July/Aug '98, p. 14), tape manufacturers—primarily TYCO and Shurtape—and UL wanted the latest LBNL tests to evaluate standard core-to-collar joints of flexible duct to



This is an example of a core-to-collar test sample showing the two taped connections and the mechanical straps.

sheet-metal collars. Their reason for changing the test joint was that the UL 181B listing is valid only for this application, and the focus of the study was to evaluate UL-listed products. Furthermore, the joint was to have no mechanical stress on it, because the Uniform Mechanical Code, sealant manufacturers' instructions, and industry guidelines from the Air Diffusion Council require separate mechanical support for ducts.

The core-to-collar duct connection consisted of a 6-inch-diameter round collar inside a flex duct core (an example is shown in photo above). Each sample had two collar-to-flex connections, one at each end of a short (about 12-inch) piece of flex duct core. One of the collars was open ended for connection to a plenum that supplied hot air. The other collar had an end cap that was internally sealed with mastic.

Four different UL 181B-listed duct tape products were tested: two

conventional duct tapes; an oriented polypropylene (OPP) acrylic adhesive tape; and a foil-backed butyl adhesive tape. A range of samples was assembled. The test samples had different numbers of wraps and/or used multiple pieces of tape for a total of 18 different combinations. A nylon strap was used to hold the connection together for 10 of the 18 samples; the other 8 samples had no strapping. For the UL listing to be valid for these tapes, they must be held in place using straps.

The test samples were exposed to heated air at 200°F and were pressurized to 0.35 inches of water (85 Pa). The exact value of this pressure is not significant, we just wanted a pressure that was within a reasonable range for residential duct systems. The surface temperature of each sample, the air temperature, and the pressure across the leaks were continuously monitored, using a data acquisition system that also controlled the temperature in the apparatus. (The top photo on p. 22 shows samples mounted on the test apparatus.)

Periodic air leakage tests and visual inspection were used to document changes in sealant performance over a two-year period. The air leakage measurements were conducted periodically (typically on a monthly or weekly basis) by removing the samples from the test apparatus.

A separate leakage-testing device pressurized the samples to 0.1 inches of water (25 Pa) and measured the air flow

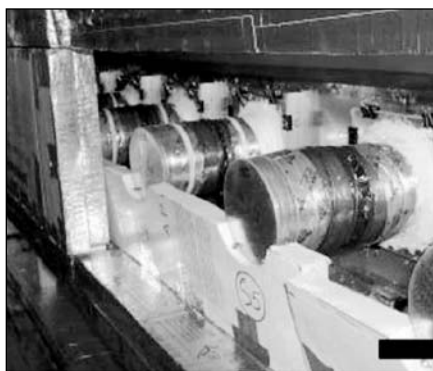
rate required to maintain this pressure difference. The failure criterion for air leakage was set at 10% of the unsealed sample leakage because this was the leakage level after which samples tended to fail rapidly in the previous testing.

A key difference between this testing and the earlier LBNL work was that in this testing there was no quantitative baseline of unsealed joint air leakage from which to objectively determine acceptable air leakage. A joint that had mechanical support including straps had relatively low leakage to begin with, so that the durability of duct tape could not be objectively measured.

Nevertheless, some interesting conclusions can be drawn from the test results.

The leakage results over the two years of testing showed no systematic increases in leakage and none of the catastrophic failures seen in our previous studies. Most of the samples showed small changes in leakage (either increases or decreases) of 0.2 CFM or less. The exception was one of the foil tape samples, which showed an increase of 0.4 CFM after the first month of testing. However, this sample then stabilized at this leakage level and did not show any significant further leakage increases. Several of the samples showed leakage reductions. Visual observations indicated that this was probably due to the flowing of the adhesive at high temperatures; the adhesive sealed more of the small cracks and leaks as it flowed.

In order to systematically record the visual deterioration of the samples, monthly pictures of all 18 specimens were taken. Typical minor deteriorations observed were discoloration, wrinkling, and oozing. Major deteriorations observed were shrinking, peeling, delamination, and cracking. Like the visual inspections of the UL 181B test, these evaluations are subjective, but they do serve to give a relative rating for each tape. Observations showed that the OPP tape had the most deterioration, while foil-butyl tape had the least deterioration. Although the OPP tape appeared to be almost disintegrated, it still maintained a good air seal. This was because the tape was not being asked to



Samples are mounted in the test apparatus.



These samples show oozing of the adhesive at the tape edges.

form a mechanical connection—only to seal the gap between the collar and the flex duct.

Strap Failure

One unexpected result of the testing was the failure of the nylon straps. Discoloration of the nylon strapping was observed within one month of the start of testing and the first strap broke after four months. Two different nylon strap

materials were used and both showed the same brittle failure. The straps usually failed at the point where the strap passes through the ratchet of the zip-tie mechanism—where the mechanical stresses are greatest.

Two straps were used—one at the cap end and one at the open end of each sample. The failure times for each end were recorded separately. The results show no significant pattern of failure for either end; the two straps on each sample generally failed within a couple of months of each other. Similarly, the failures were independent of the tape they were used with. All but two straps had failed by the end of our testing. The two remaining straps showed the same discoloration as the failed straps, but they had not broken and fallen off. Strap failure is a major problem, because mechanical attachment thereafter is maintained only by the duct sealant. If ducts are not well supported, significant mechanical stress can occur to cause the sealant to fail after the strap fails. In extreme cases, the duct connection may separate.

The materials used for the straps were typical of those used in the field, which have an unknown temperature rating. Product literature from strap manufacturers shows that there are other strap materials that have higher temperature ratings—such as Heat Stabilized Nylon 6% for continuous exposure above 185°F and TEFZEL for even higher temperatures. Straps made of these materials may have improved high-temperature durability. As an alternative, the authors recommend metal straps, because they have no temperature degradation.

UL181B Revisions for Straps

In 2003, perhaps in response to LBNL's preliminary reports, UL 181B was revised to include a new set of tests for fasteners, including straps. Products that pass this set of tests are to be marked UL 181B-C. Because LBNL testing started before the revisions for straps were added, the researchers were not able to evaluate the performance of UL-listed straps. The tests cover tensile

strength, smoke spread, heat production, mold growth, tension (mechanical integrity of the connection), air leakage, and low- and high-temperature aging.

The most relevant test for longevity is the one done under high temperatures. In this test, the straps are heated to 212°F for 60 days. The straps are tested for tensile strength before and after they undergo heating; they must retain 75% of their initial strength after they have been heated for 60 days. The tensile testing itself is not conducted at high temperature; the straps are conditioned for 48 hours at 73°F (23°C) and 50% relative humidity (RH) and are then tensile tested at some unspecified temperature. In other words, their performance at elevated temperature is not evaluated. The object of the UL test is to learn how exposure to high temperature affects the properties of the material.

The new UL test is a good step forward, but it falls short of being a good indicator of strap performance. There are several reasons why:

- The straps are not tested in the failure mode that is observed, such as brittle failure in bending. We do not know if the strapping materials show greater or lower tensile strength as they become more brittle; many ductile materials, in fact, show greater tensile strength as they become brittle. In addition, the straps are not subjected to any strain during the UL high-temperature baking. In real applications (and in laboratory testing), the straps are under the influence of combined heat and strain. Without additional testing, we do not know if strain is a factor in failure, but in general, we would expect that it is.

- The testing is for 60 days only. In the LBNL study, none of the straps failed in 60 days. Testing for only 60 days appears to be insufficient. Given the relatively arbitrary nature of selecting time limit criteria, we should select an appropriate time limit that allows us to differentiate between acceptable and non-acceptable performance.

- The wording of the test is somewhat unclear. This makes it extremely difficult to know whether one is following the testing procedure exactly.



The cloth-backed tape sample shows considerable shrinkage and delamination.



The OPP tape showed considerable visual degradation, even though it still provided a good air seal.

New ASTM Standard

Because the UL tests do not sufficiently address durability issues, researchers have worked with ASTM to develop a new standard: ASTM E2342-03 "Standard Test Method for Durability Testing of Duct Sealants." This standard tests sealants on a collar-to-plenum connection. This

connection is much harder for tapes to seal, because the tape must conform to a complex three-dimensional shape, and because this shape mechanically stresses the tape. In the future we hope that the ASTM standard will be used to specify durable duct sealants, so that builders will not have to rely on the UL listing, which does not sufficiently address durability.

Recommendations for Better Duct Sealing

Our recent research suggests that almost any tape can be made to work for a core-to-collar joint when perfectly applied. Given all the better choices available and all the problems that can and do occur in the field, we would not recommend that cloth-backed rubber adhesive tapes be used, unless they can pass a suitable E2342 test.

If visual degradation is an important factor, OPP and cloth-backed tapes should be avoided. While tapes that fail usually show visual degradation, the converse is not always true. We speculate, however, that visually degraded tapes are more likely to be susceptible to damage from vibration and other mechanical stresses.

Conventional nylon straps are quite prone to fail at higher temperatures, but no systematic studies have yet been done across product classes. UL has proposed a new standard for straps, but the protocol for the current UL test does not suggest that it would be a good indicator of durability. Products meeting the new UL standard for straps should be independently tested for the failure modes observed in the field and laboratory to determine if that test is a reasonable indicator of durability.

Until a suitable test for strap durability exists, we recommend that only high-temperature nylon or metal straps be used—especially in cases where tapes may be sensitive to mechanical stresses.

Iain Walker is **Home Energy's** executive editor. Iain Walker and Max Sherman are staff scientists in the Energy Performance of Buildings Group at Lawrence Berkeley National Laboratory in Berkeley, California.